GroundWalks (WalkSensePlace)

Reading the City Through Your Feet to Design Healthier Urban Places

Kristian **Kloeckl** (CAMD, urban interaction design)
Eric **Folmar** (Bouvé, physiotherapy and biomechanics)
Amy **Lu** (CAMD/Bouvé, health tech and communication)
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"Walking is man's best medicine."

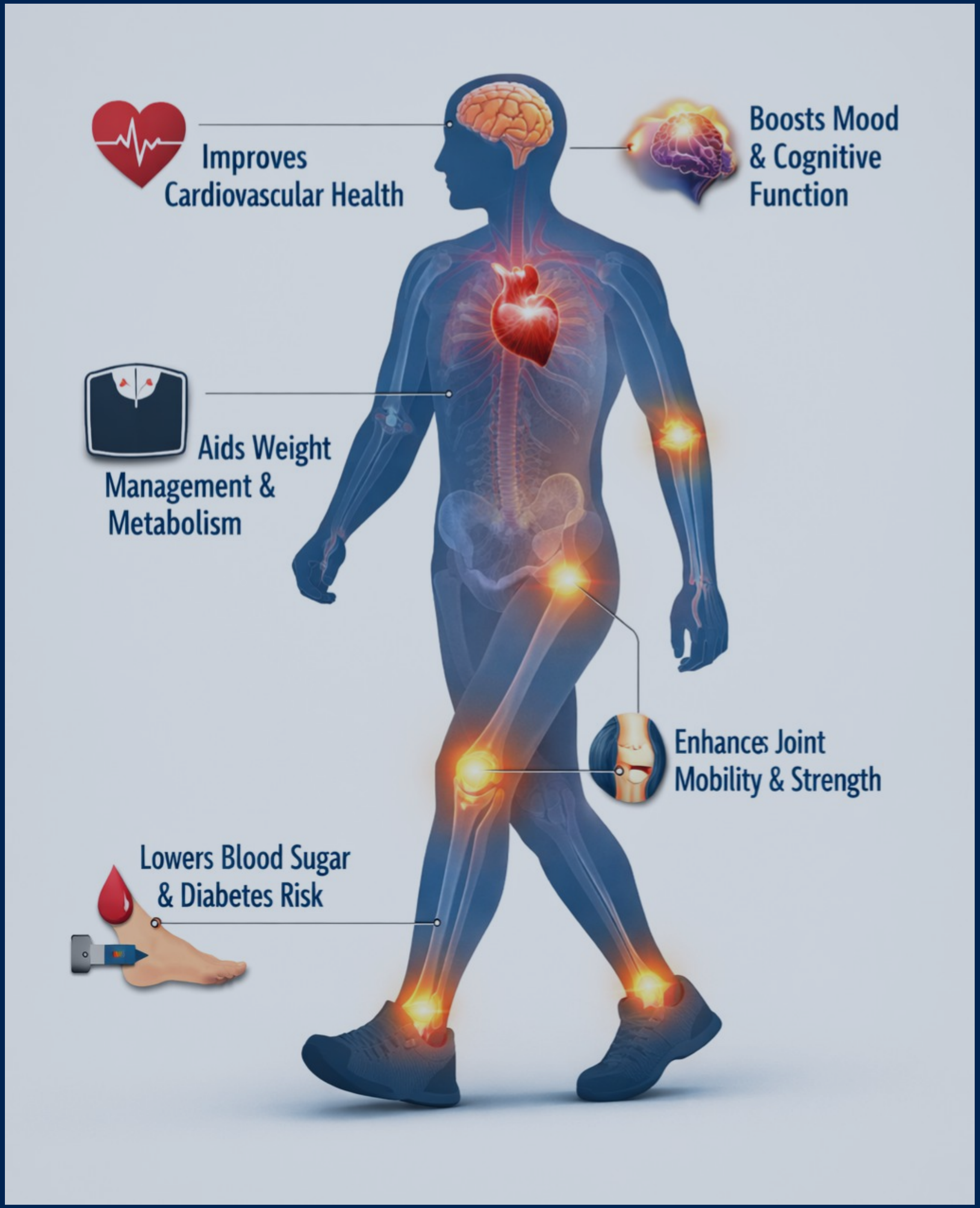
— *Hippocrates*

"The rhythm of walking generates a kind of rhythm of thinking, …”

— *Rebecca Solnit, Wanderlust*

Walking…

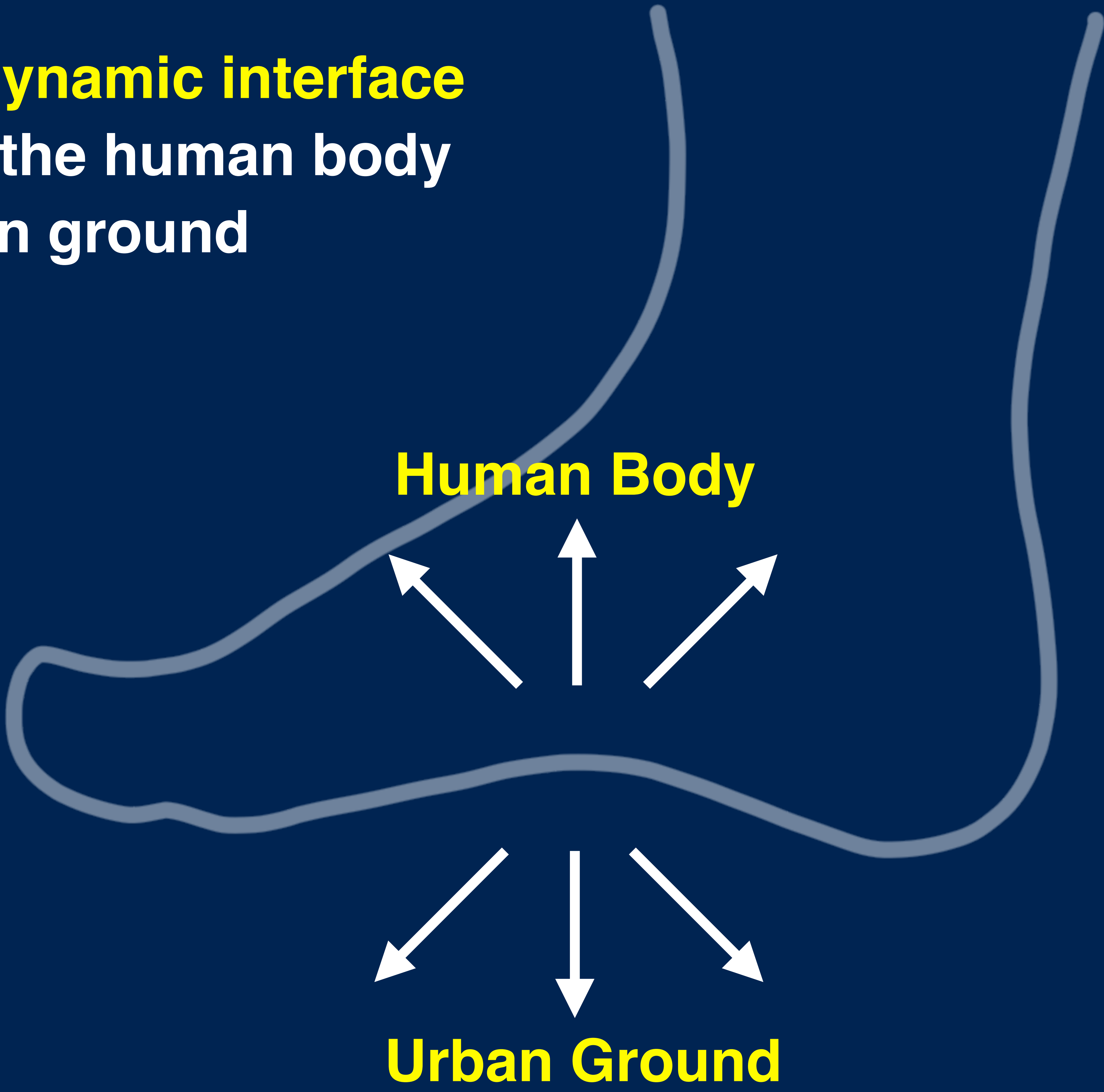
One of the most **accessible and sustainable forms of physical activity**. No specialized equipment and minimal training required.



Walkable Cities Movement
& Public Health

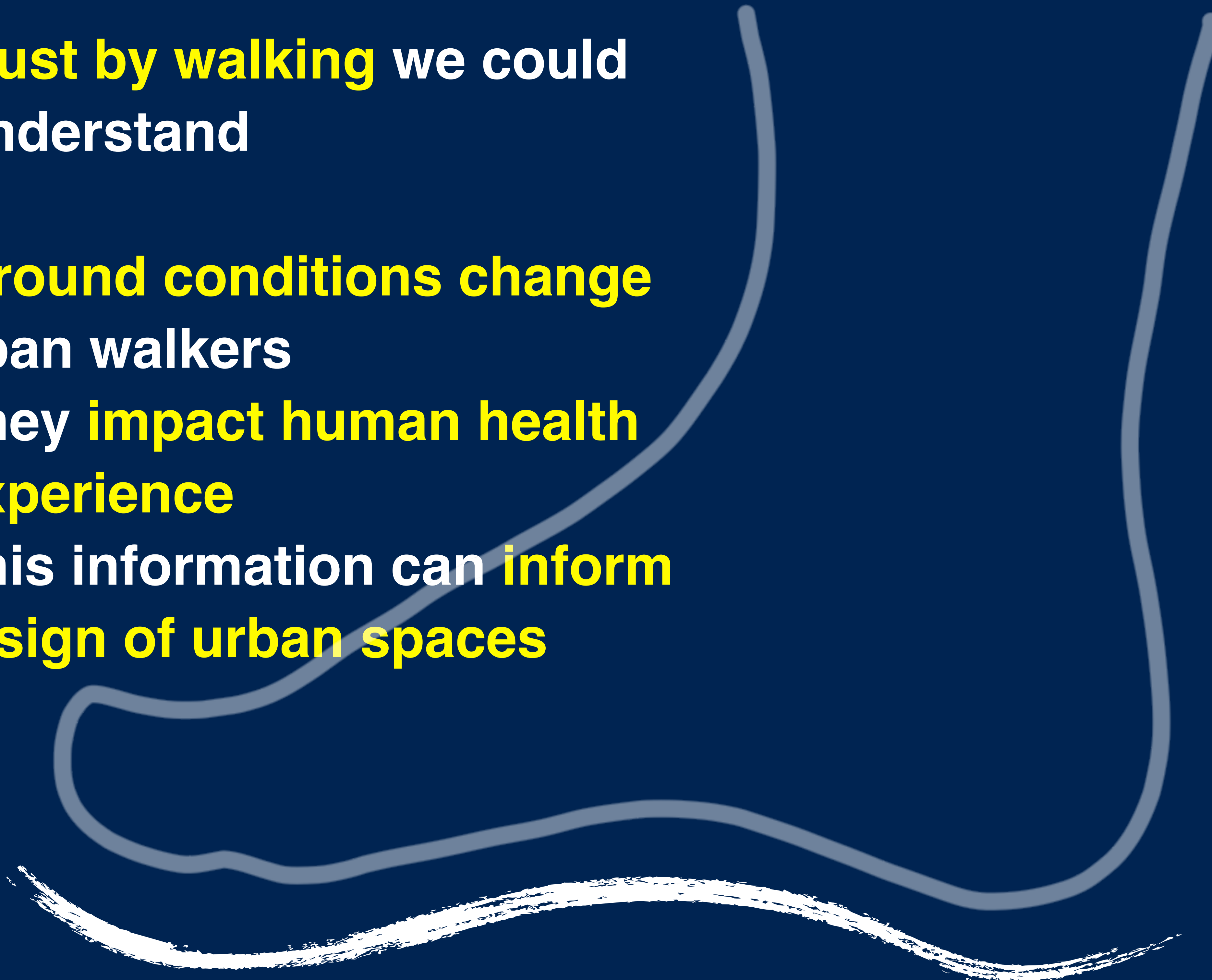
Walking can be integrated into daily routines.
Even moderate increases in walking can yield significant health gains at the population level.

Feet as dynamic interface between the human body and urban ground

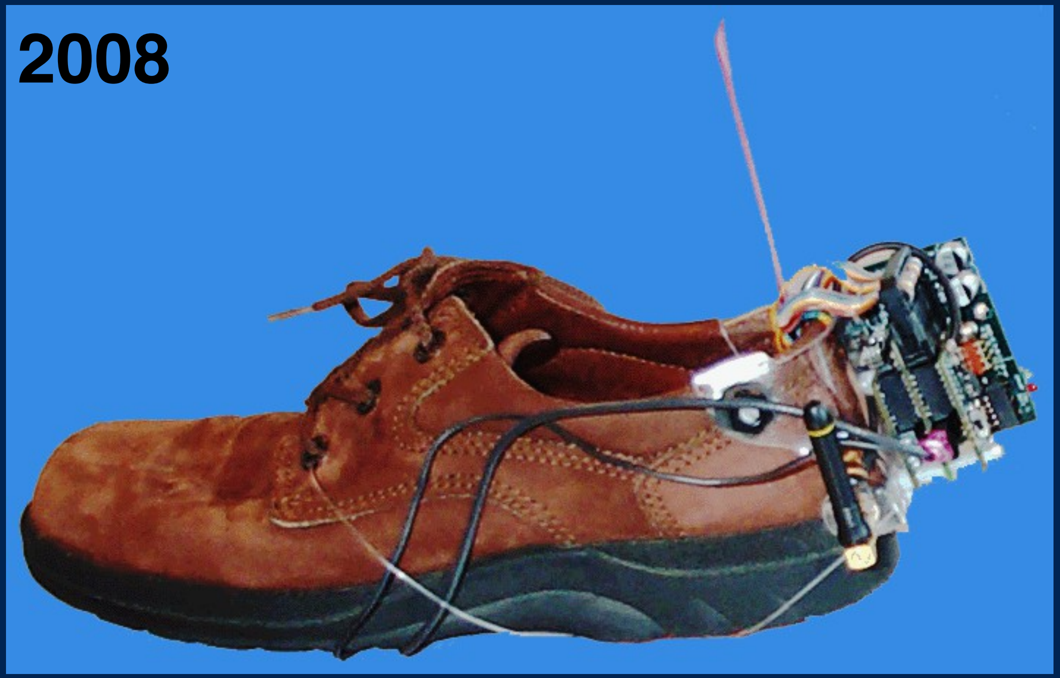


What if **just by walking** we could better understand

- how **ground conditions change** for urban walkers
- how they **impact human health and experience**
- how this information can **inform the design of urban spaces**

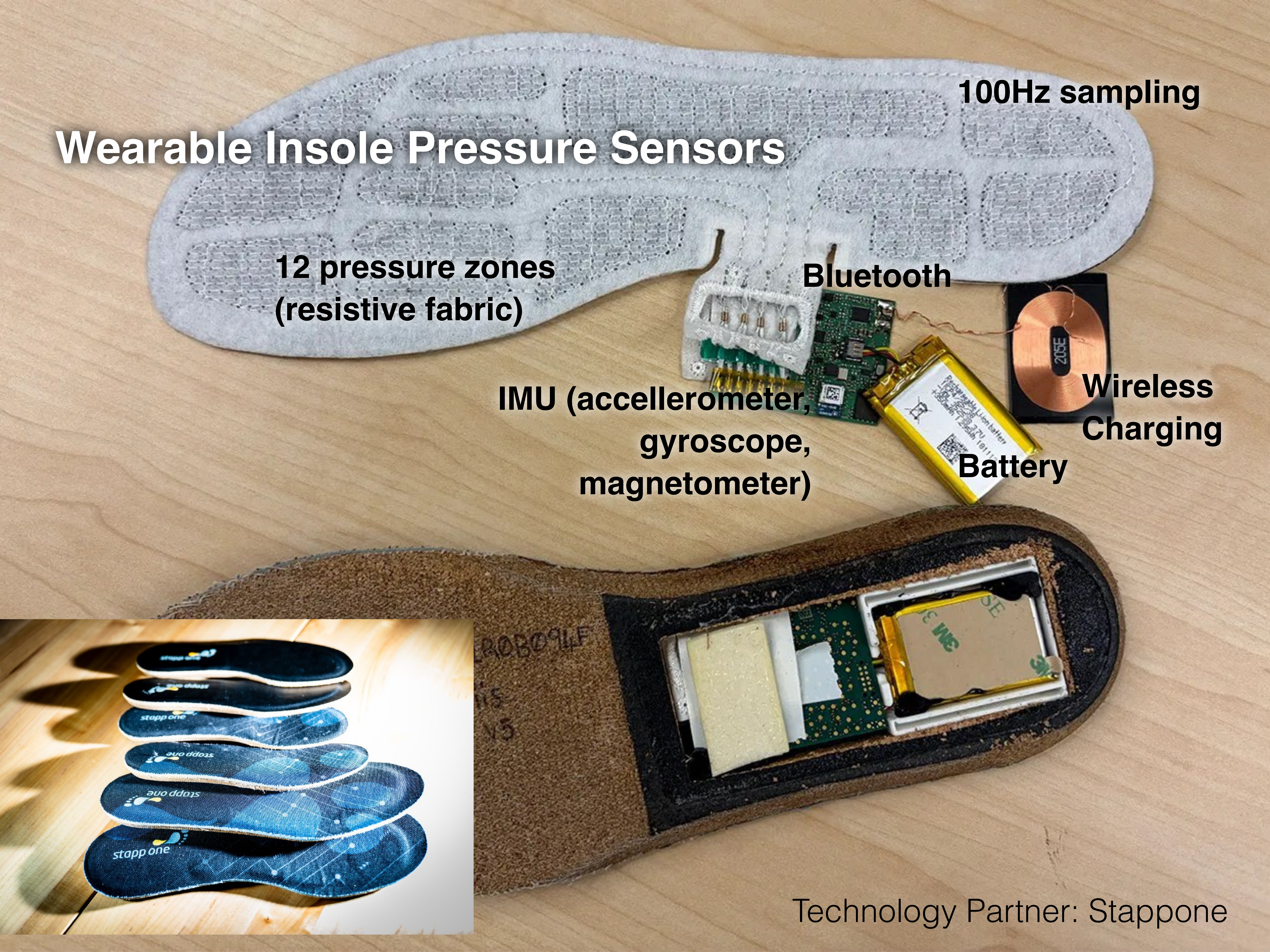


Wearable Insole Pressure Sensors

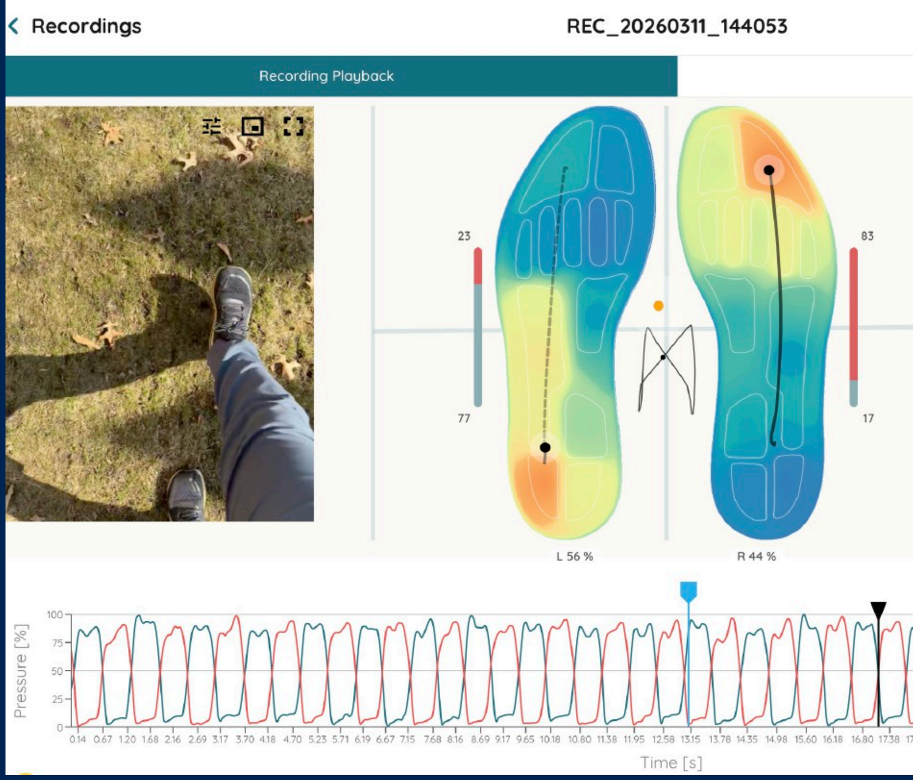


Capacitive Pressure Sensors. All other components are external

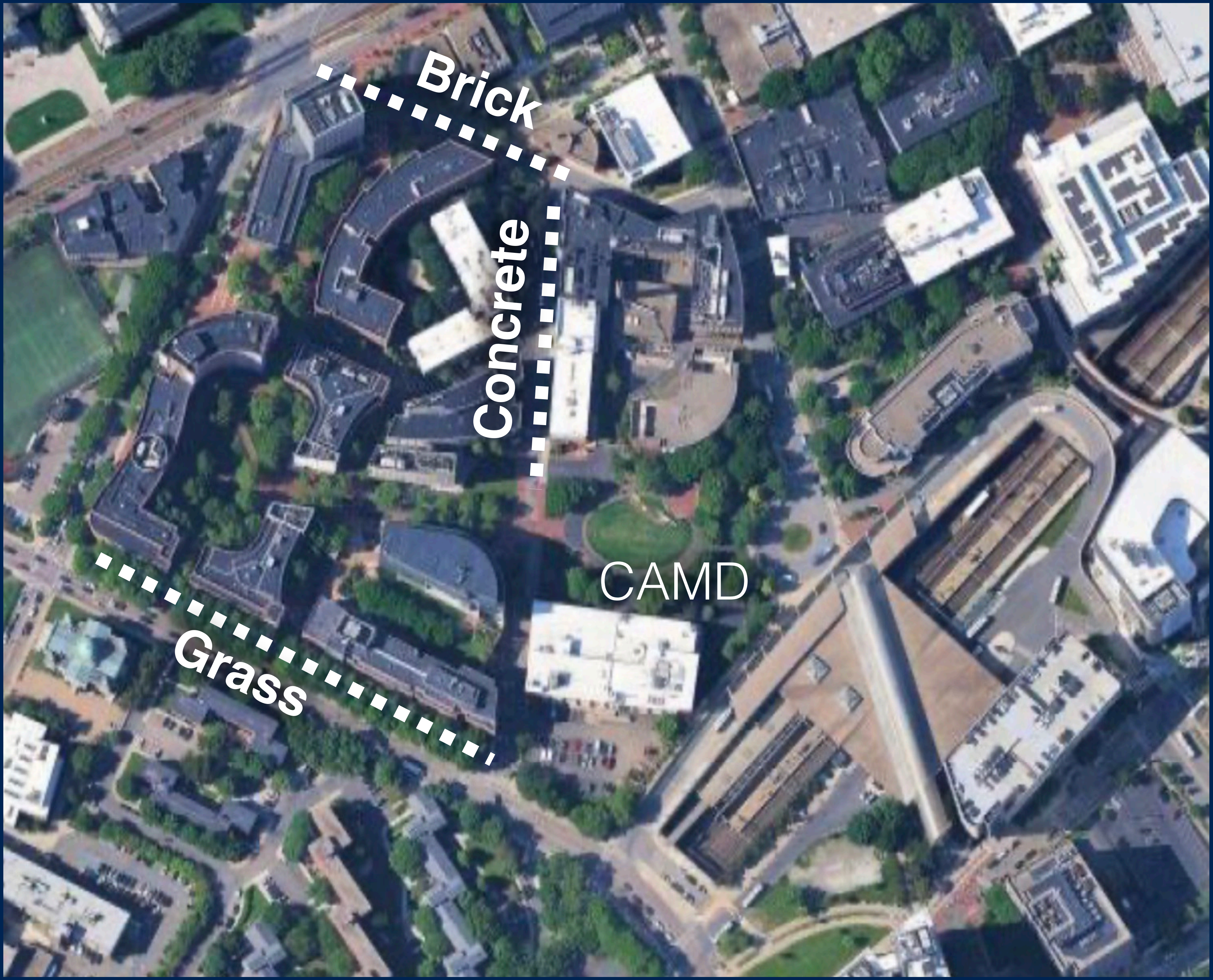
Inductive Fabric Pressure Sensors Integrated Battery, IMU, Wireless Charging, Bluetooth.


Wearable Insole Pressure Sensors
100Hz sampling
12 pressure zones
(resistive fabric)
Bluetooth
IMU (accellerometer,
gyroscope,
magnetometer)
Wireless
Charging
Battery
Technology Partner: Stappone

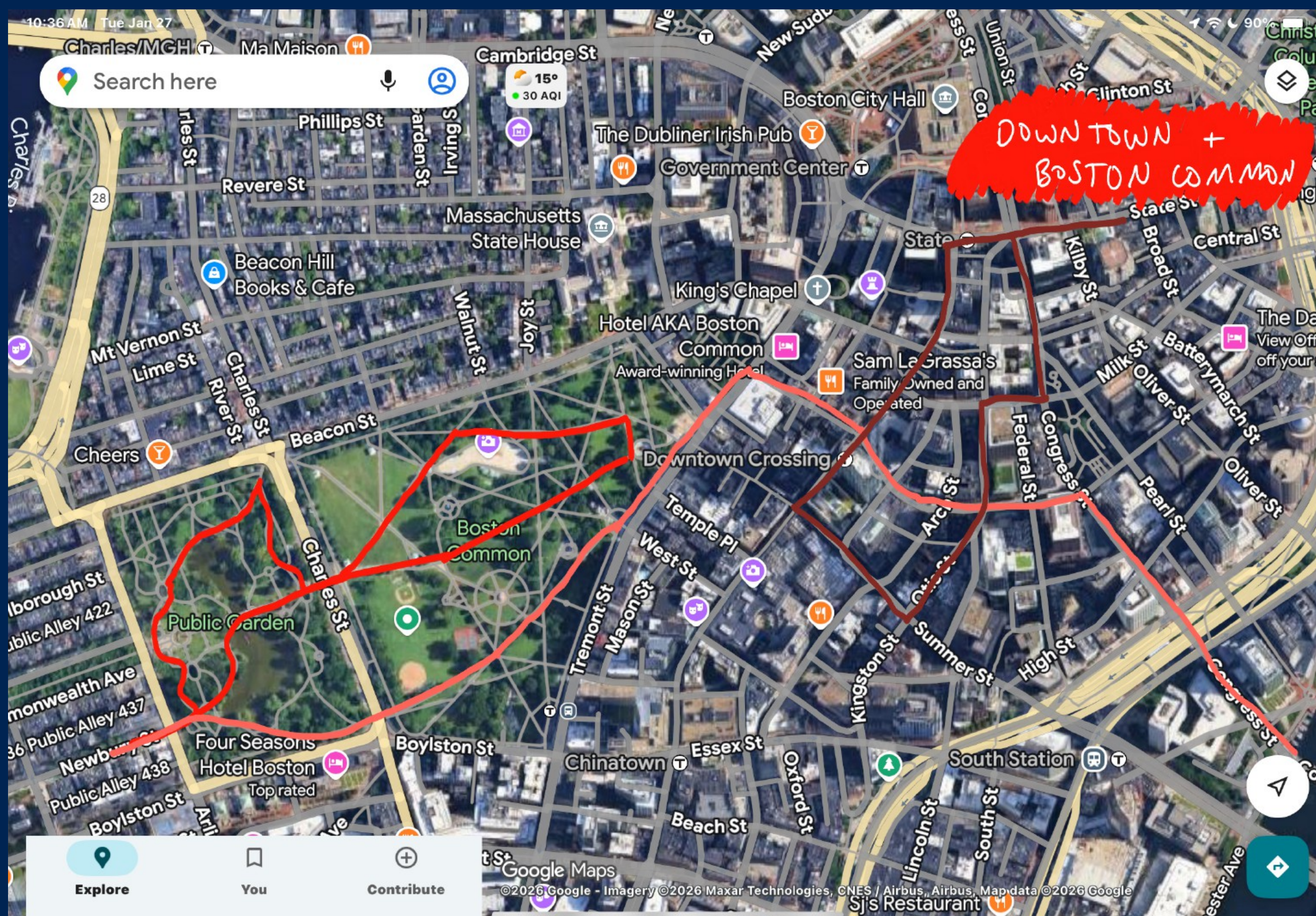
Data Collection while walking
on different ground conditions

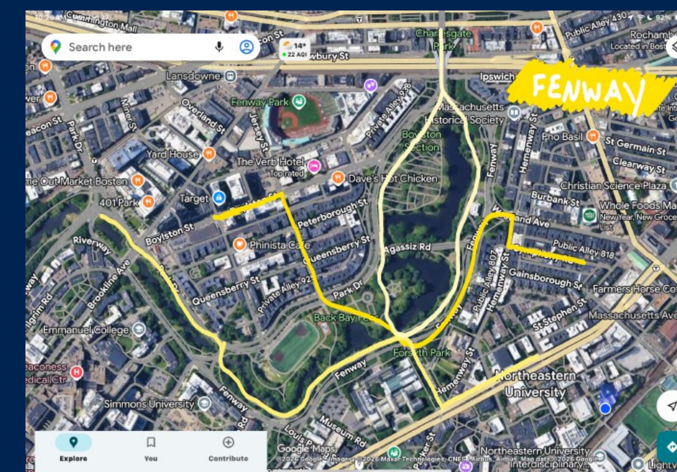


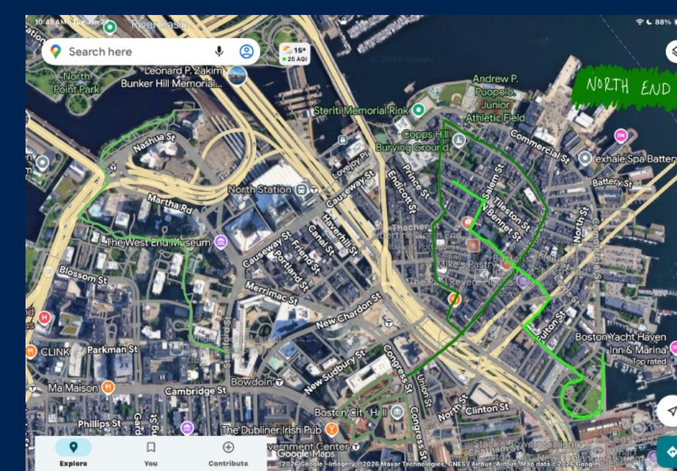
Test routes for diverse ground conditions

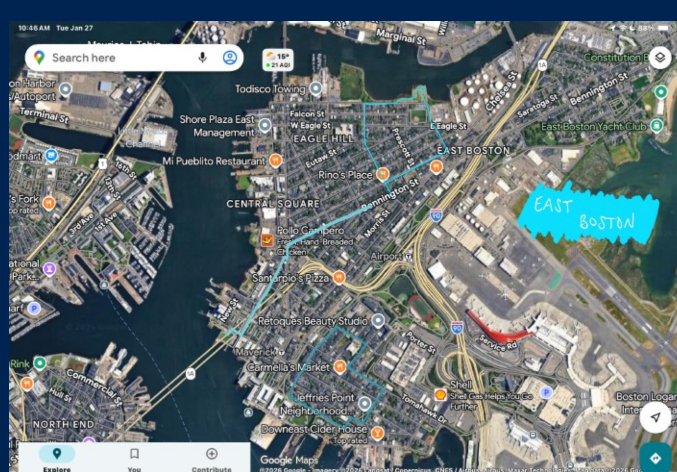


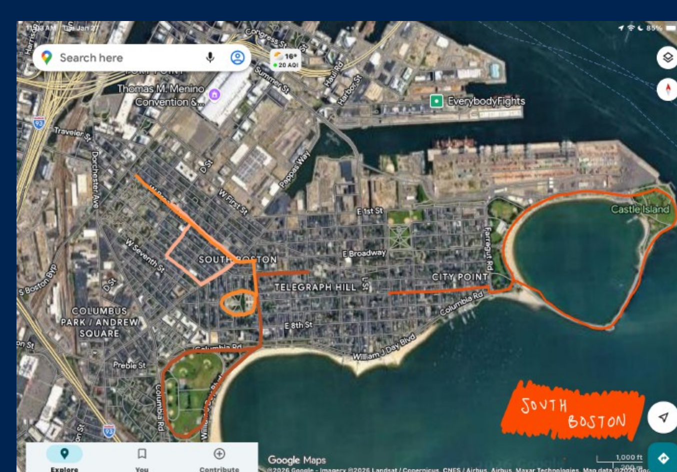
Interviews with urban walkers about walking routes

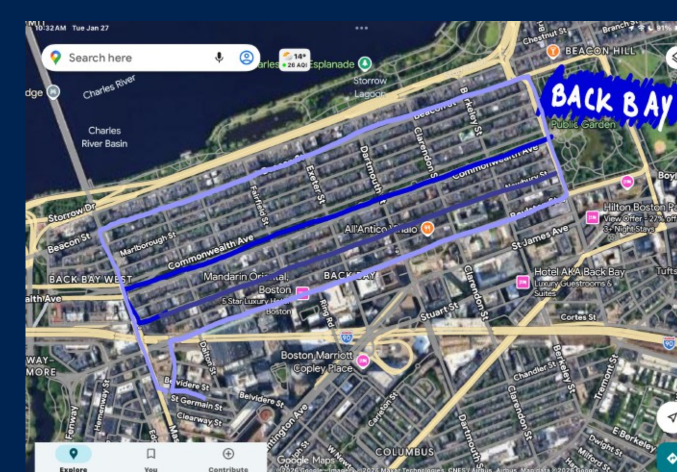




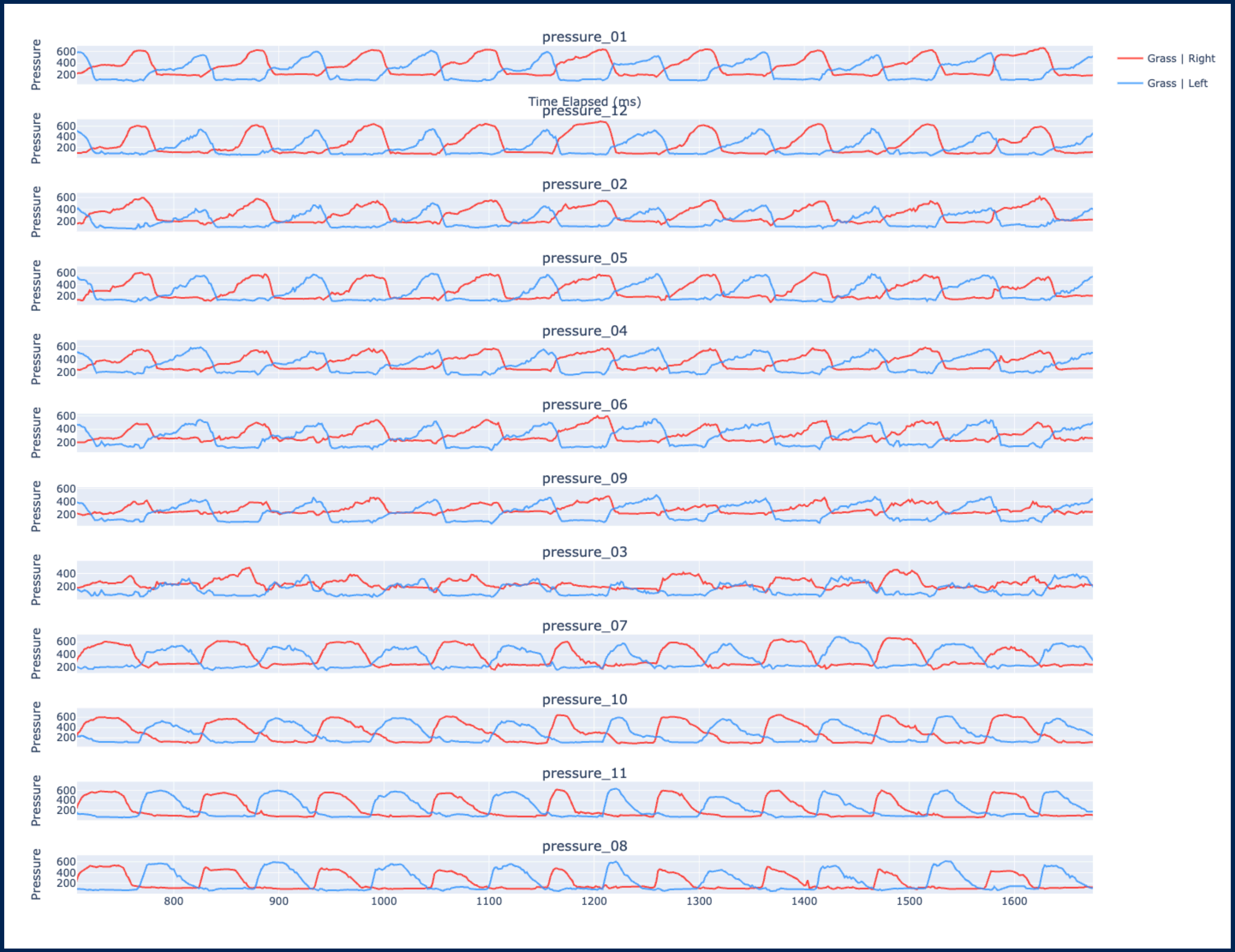




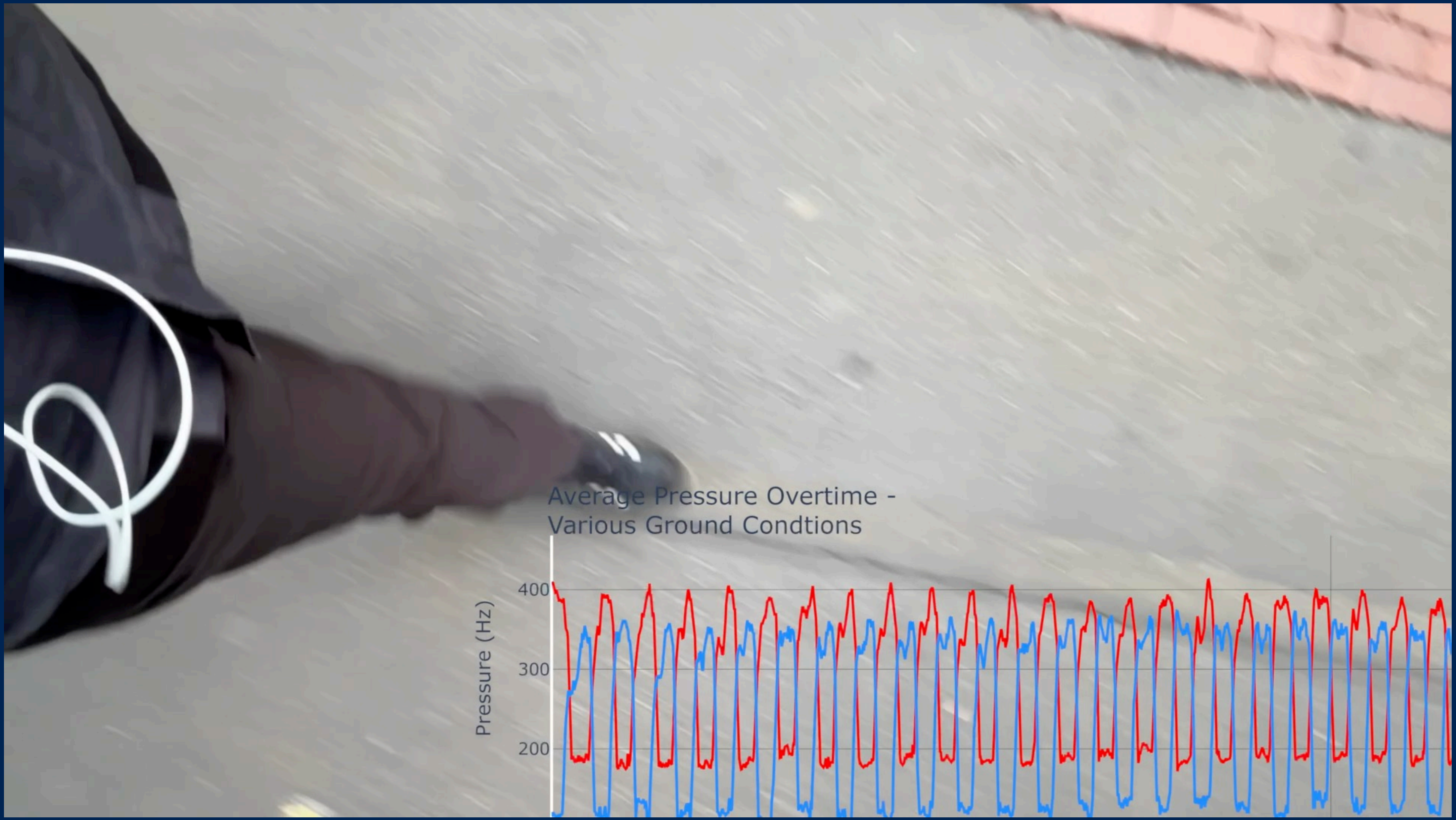




Data visualizations | Gait Cycle Time Series

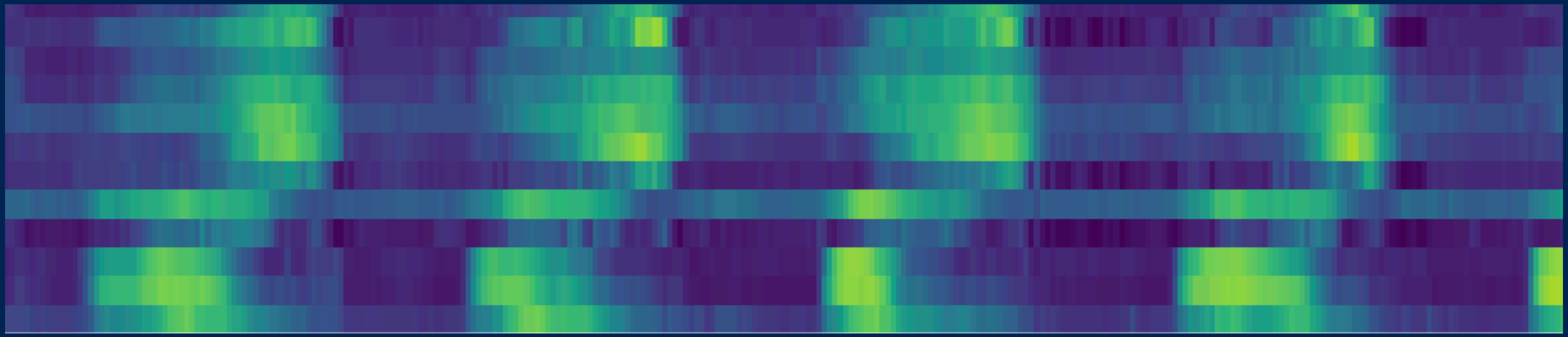


Data visualizations | Gait Cycle Time Series

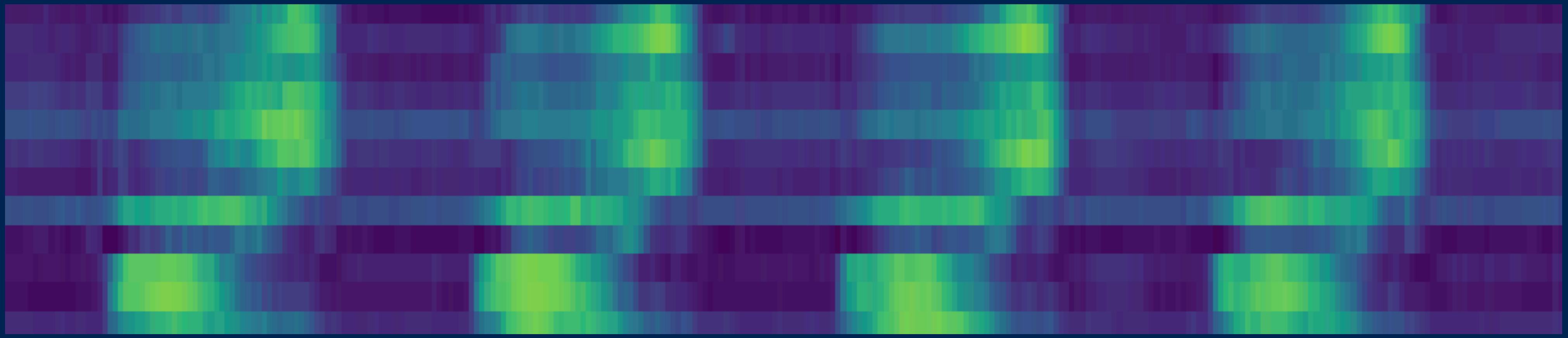


Data visualizations | Pressure Sensor Heatmaps

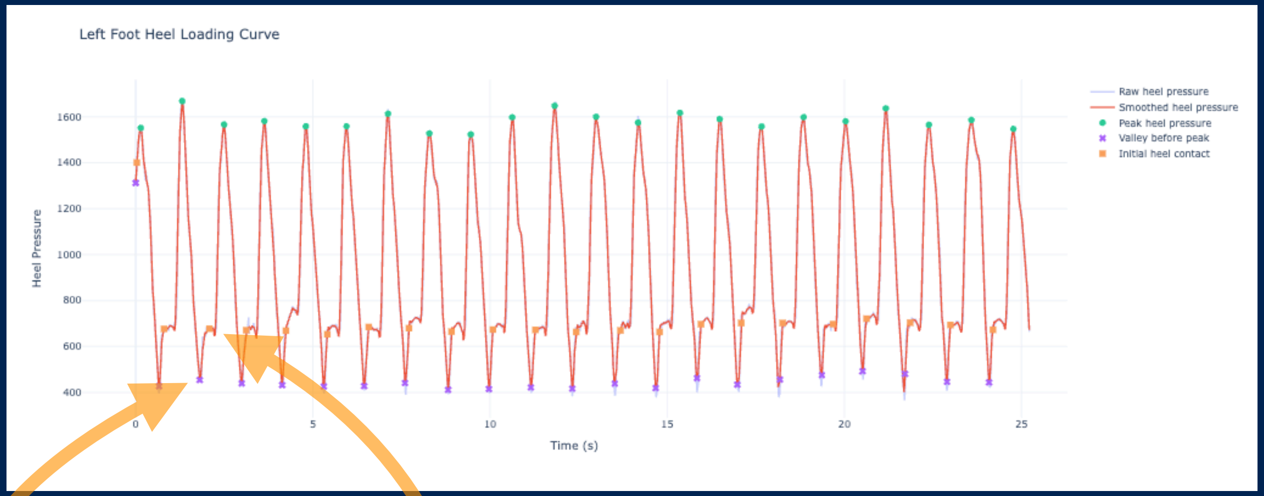
Asphalt

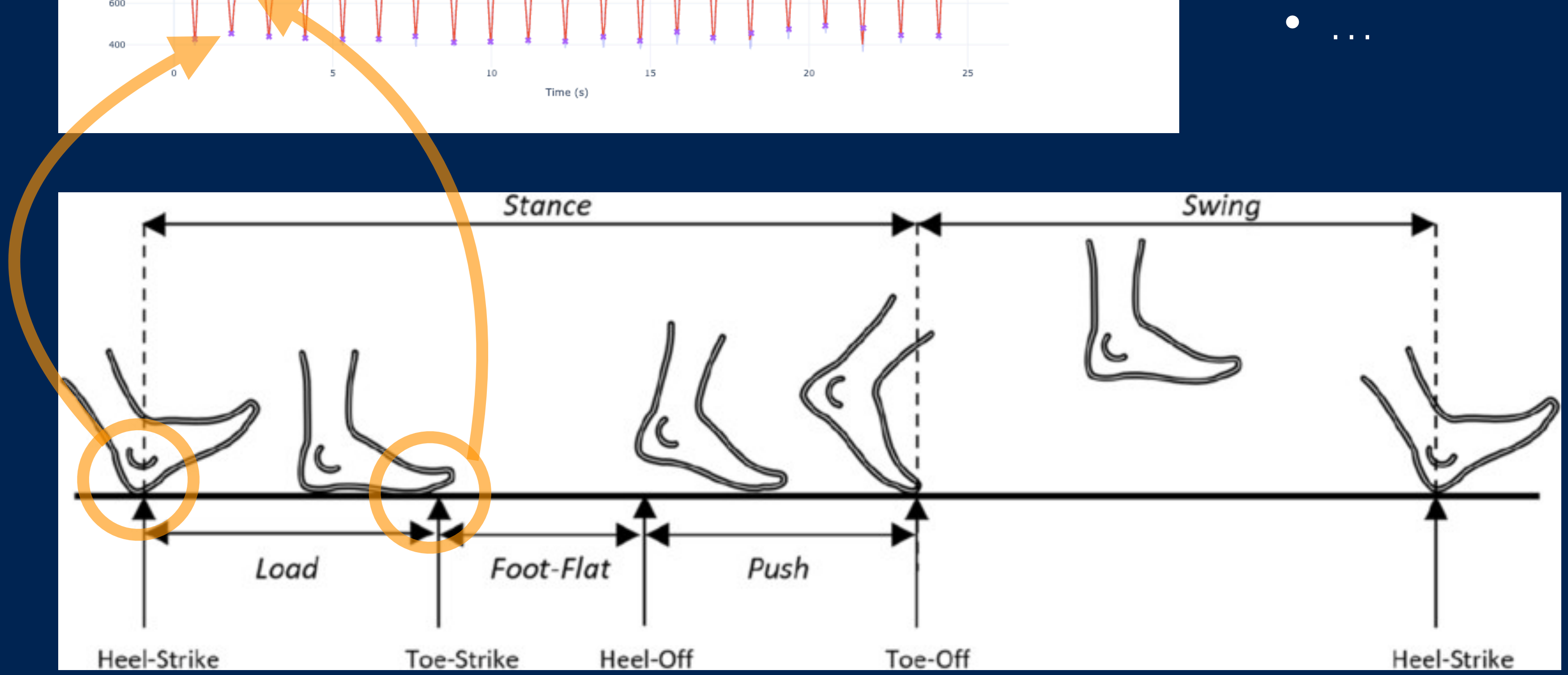


Grass



Data segmentation / Features / Machine learning

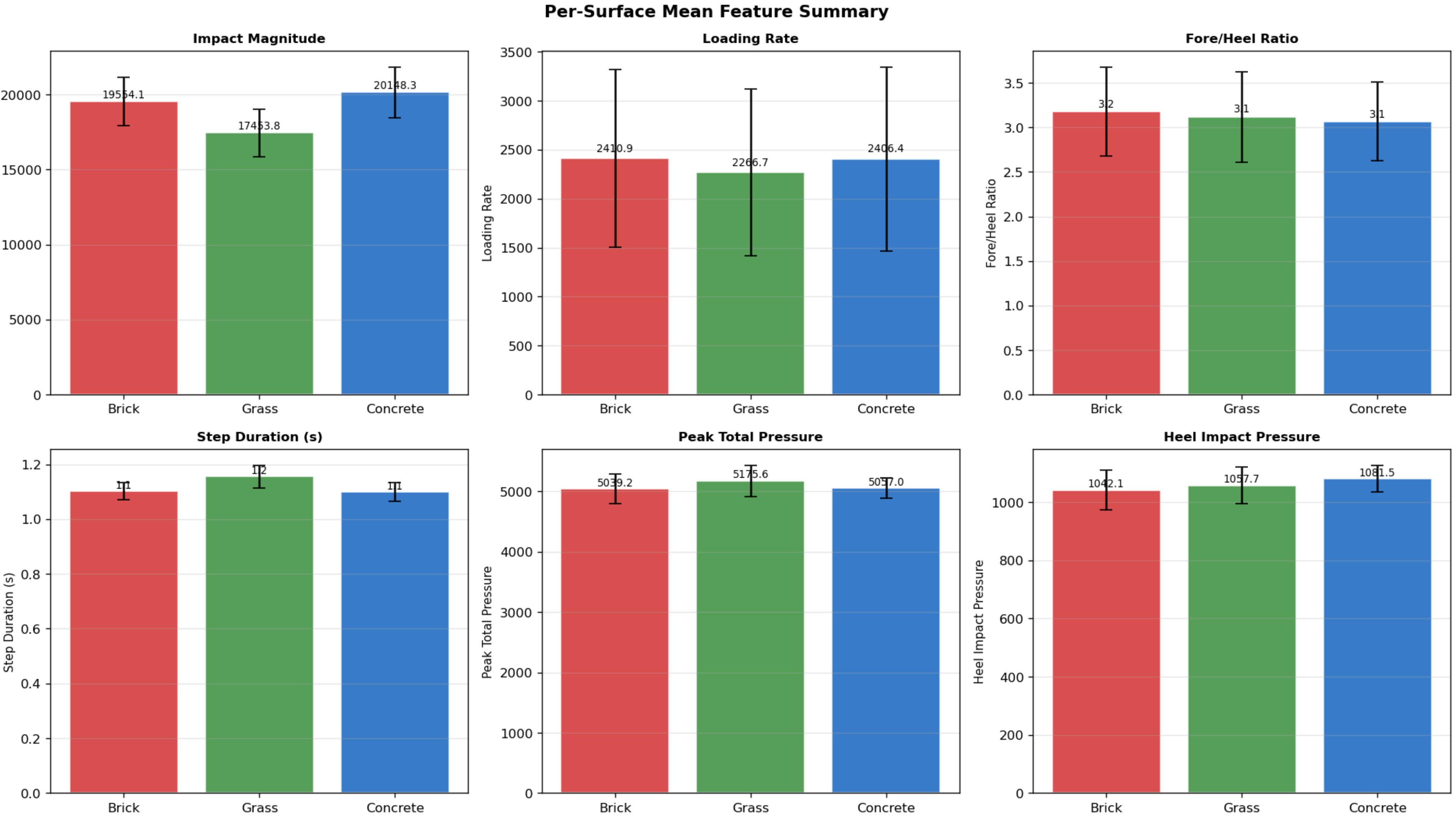


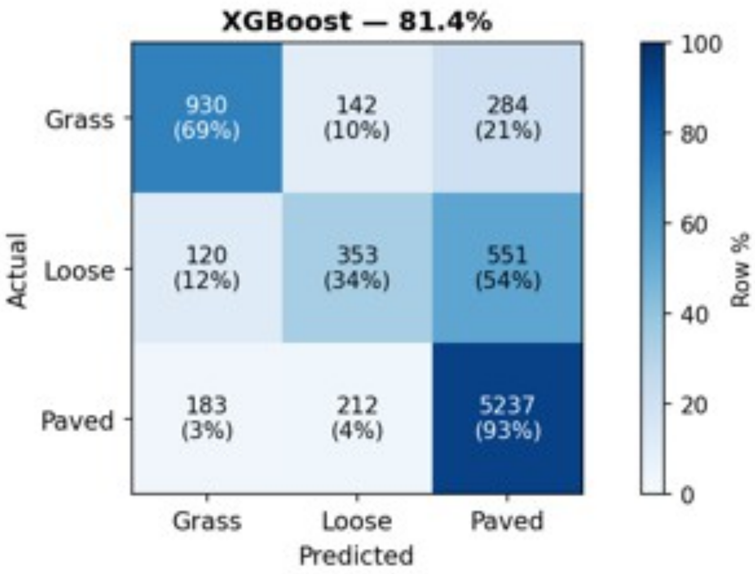


Feature Constructing at nexus of Physiotherapy, Urban Design and Public Health:

- Accelerometer Impact Peak at Heel Strike
- Spectral Energy Ratio (High vs Low Frequency)
- Loading Rate
- Pressure Distribution Entropy During Mid-Stance
- Step-to-Step Variability of Feature
- …

Data segmentation / Features / Machine learning

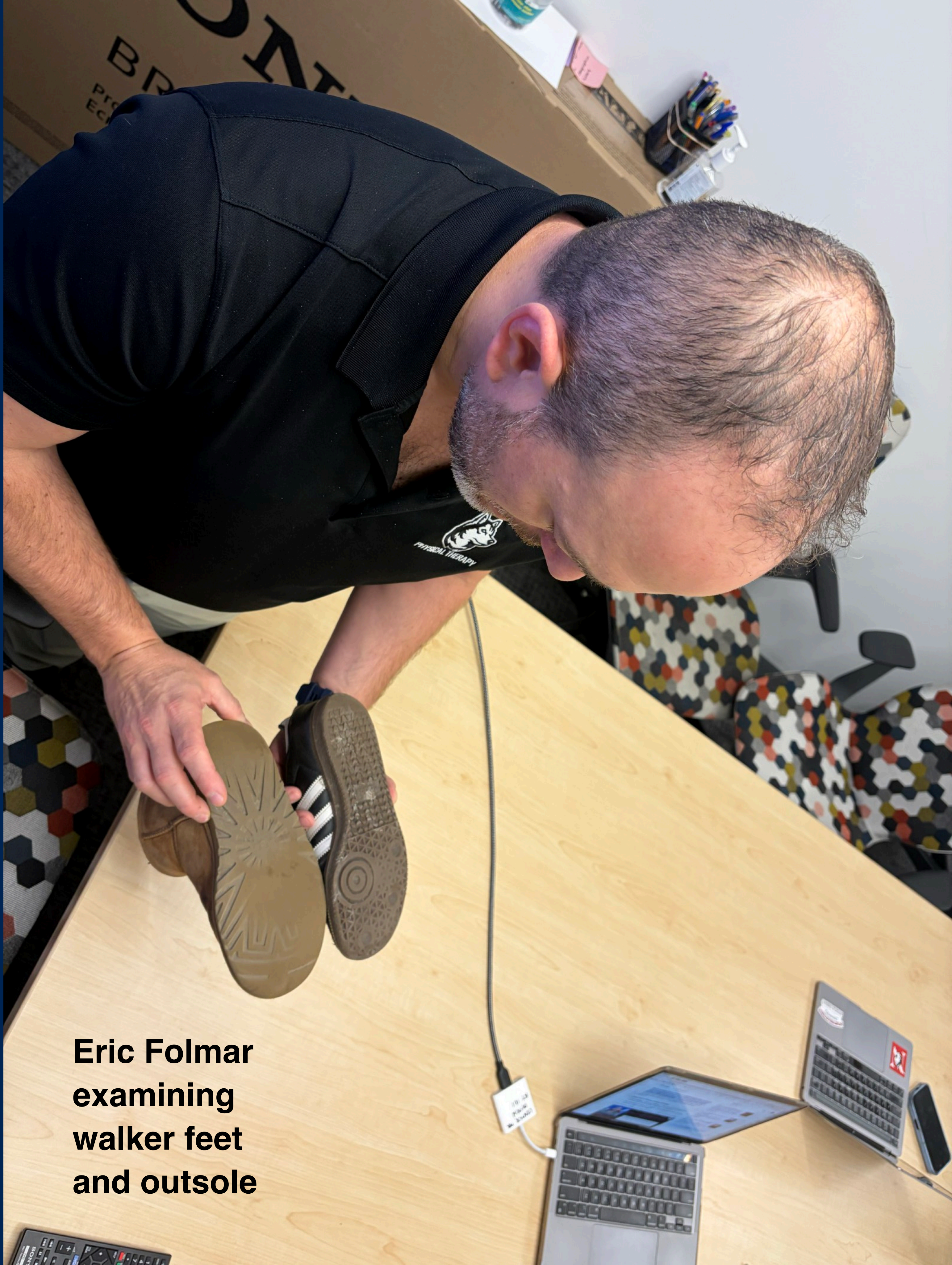




Data Traces

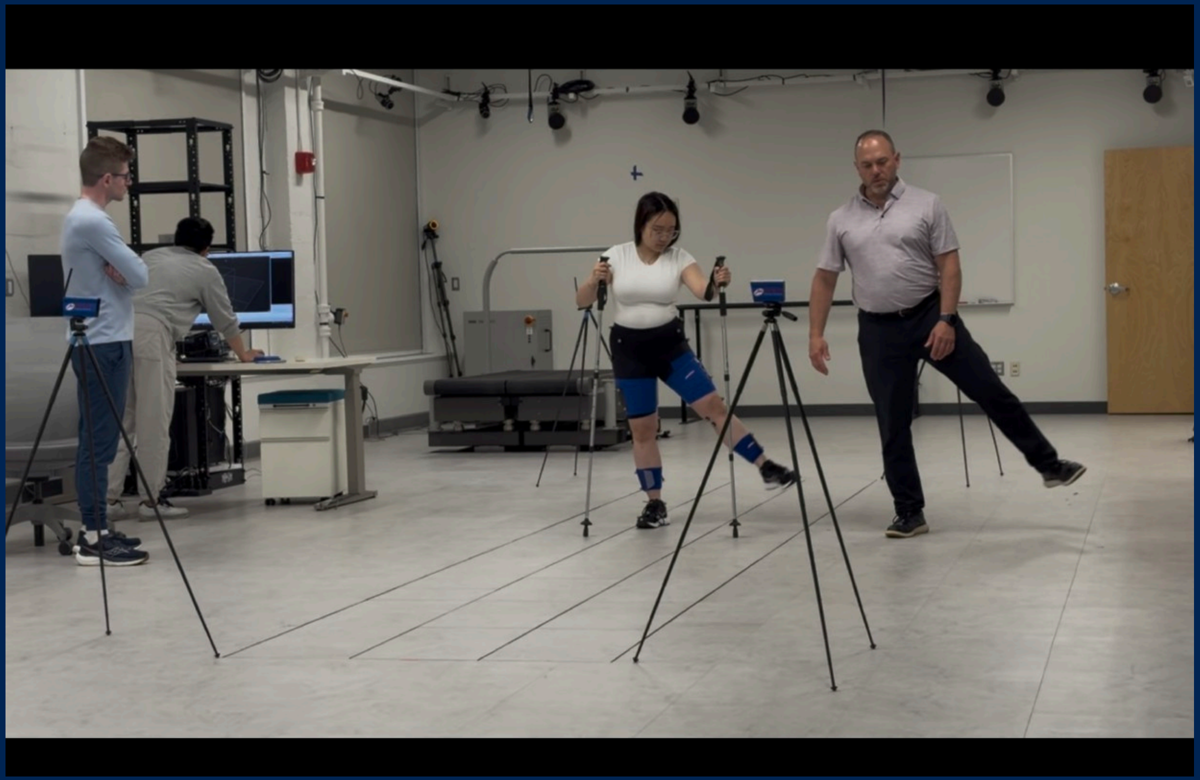
Foot and biomechanics assessment as ground truth

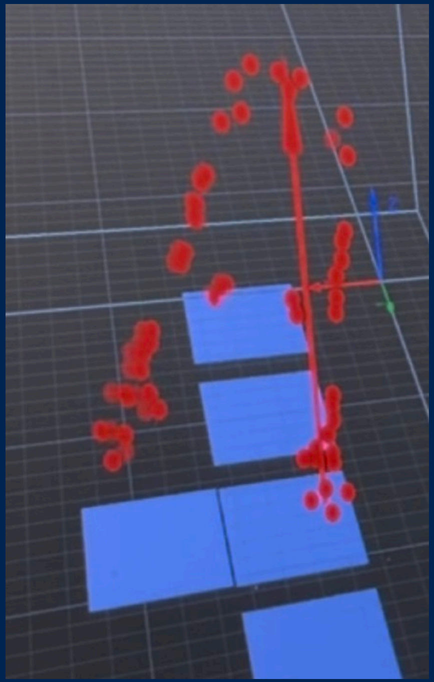
Data traces mark shoe soles based on walking type



Eric Folmar examining walker feet and outsole

Biomechanics Lab Data Collection





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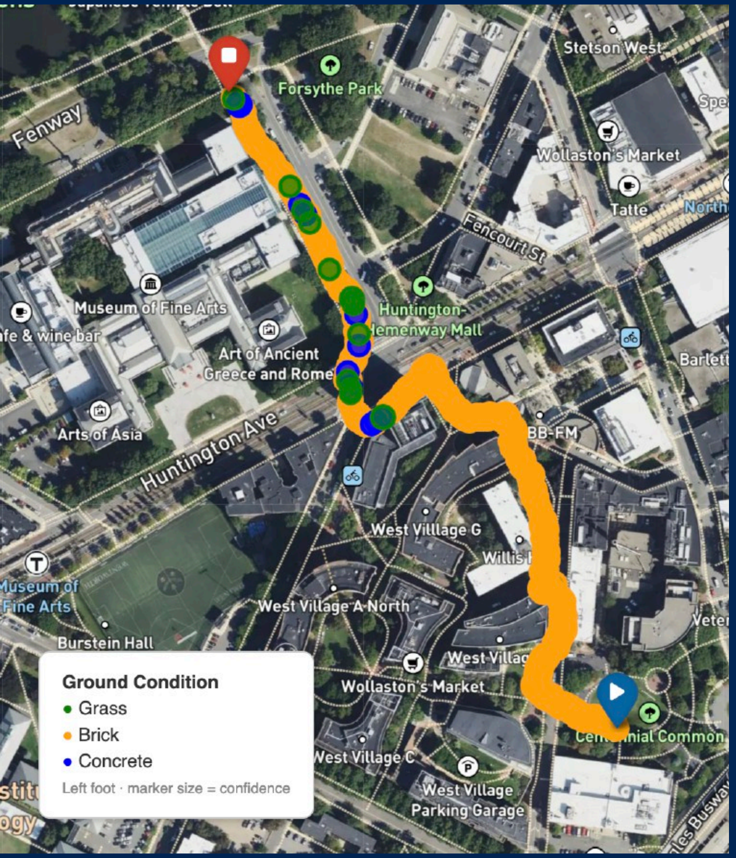

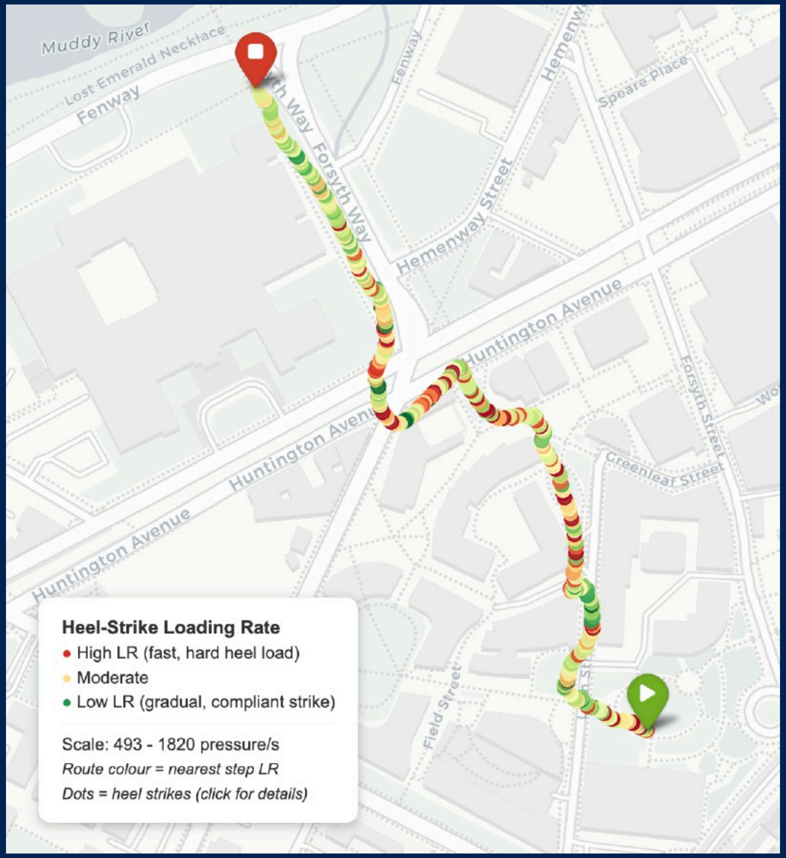

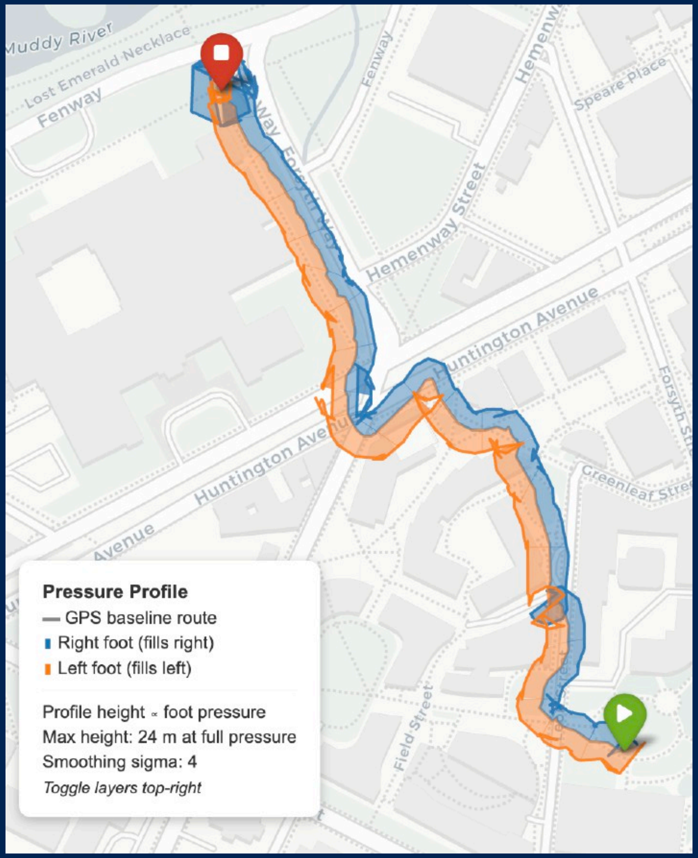

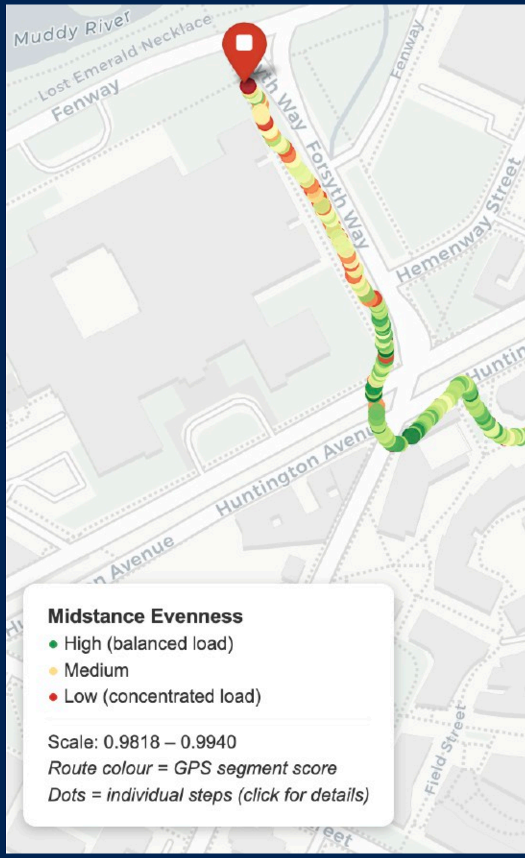

Next

From number of steps to **quality of steps.**

Map different **neighborhoods** by what **ground conditions** they afford for walking.

Develop framework to integrate this research in **design of walkable cities.**

How **different grounds** reinforce or complement a person's **walking type**.

Outreach:

- Pedestrian advocacy/research organization
- Shoe manufacturer

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